

T05 — Geometry validation · all metrics pass criteria

Asn A 39 Ramachandran outlier confirmed by 3 independent code paths · agent’s “0.00%” rounds 1 outlier to zero

Metric	Tool	Value	Verdict
Clashscore (cctbx)	mmtbx.validation_summary	3.13	✓ < 5
Clashscore (non-cctbx)	MolProbity standalone	3.63	✓ < 5
MolProbity composite	mmtbx.validation_summary	1.63	✓ < 2
Ramachandran outliers	mmtbx.validation_summary	0.53% (Asn A 39)	✓ <1% · headline wrong
Ramachandran favored	mmtbx.validation_summary	98.40%	✓ > 98%
Rotamer outliers	mmtbx.validation_summary	4.88%	✓ < 5%
C-β deviations	mmtbx.validation_summary	0	✓
Bond RMSZ (CCP4 dict)	REFMAC5	0.884 σ	✓ < 1 σ
Angle RMSZ (CCP4 dict)	REFMAC5	0.864 σ	✓ < 1 σ
Holton geom-energy ratio	phenix.holton_geometry_validation	1.33 σ (start 3.32)	✓

Asn A 39 — Ramachandran outlier confirmed by 3 code paths

mmtbx.validation_summary

0.53% outliers · 1 residue

phenix.holton_geometry_validation

Asn A 39 flagged as outlier

ProSMART (non-cctbx)

Procrustes 0.526 vs avg 0.27–0.40 · 12° hinge · MaxDist 1.31 Å